

```

import java.rmi.Naming;
import java.util.Scanner;

public class ArithmeticClient {
    public static void main(String[] args) {
        try {
            ArithmeticService arithmeticService =
                (ArithmeticService) Naming.lookup(
                    "rmi://localhost/ArithmeticService"
                );

            Scanner scanner = new Scanner(System.in);

            System.out.print("Enter the first number: ");
            double num1 = scanner.nextDouble();

            System.out.print("Enter the second number: ");
            double num2 = scanner.nextDouble();

            System.out.println("\nChoose an operation:");
            System.out.println("1. Add");
            System.out.println("2. Subtract");
            System.out.println("3. Multiply");
            System.out.println("4. Divide");
            System.out.print("Enter your choice (1-4): ");
            int choice = scanner.nextInt();

            double result = 0;
            switch (choice) {
                case 1: result = arithmeticService.add(num1, num2); break;
                case 2: result = arithmeticService.subtract(num1, num2); break;
                case 3: result = arithmeticService.multiply(num1, num2); break;
                case 4: result = arithmeticService.divide(num1, num2); break;
                default:
                    System.out.println("Invalid choice");
                    return;
            }

            System.out.println("Result: " + result);
            scanner.close();

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```


```

import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;

public class ArithmeticServer {
    public static void main(String[] args) {
        try {
            // Start RMI registry on default port 1099
            LocateRegistry.createRegistry(1099);

            ArithmeticServiceImpl arithmeticService = new ArithmeticServiceImpl();

            // Bind service
            Naming.rebind("rmi://localhost/ArithmeticService", arithmeticService);

            System.out.println("Arithmetic Server is ready...");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```

```

-----
import java.rmi.Remote;
import java.rmi.RemoteException;

```

```

public interface ArithmeticService extends Remote {
    double add(double a, double b) throws RemoteException;
    double subtract(double a, double b) throws RemoteException;
    double multiply(double a, double b) throws RemoteException;
    double divide(double a, double b) throws RemoteException;
}

```

```

-----
// Source code is decompiled from a .class file using FernFlower decompiler (from
IntelliJ IDEA).

```

```

import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;

public class ArithmeticServiceImpl extends UnicastRemoteObject implements
ArithmeticService {
    public ArithmeticServiceImpl() throws RemoteException {
    }

    public double add(double var1, double var3) throws RemoteException {
        return var1 + var3;
    }
}

```

```
public double subtract(double var1, double var3) throws RemoteException {
    return var1 - var3;
}

public double multiply(double var1, double var3) throws RemoteException {
    return var1 * var3;
}

public double divide(double var1, double var3) throws RemoteException {
    if (var3 == 0.0) {
        throw new RemoteException("Cannot divide by zero");
    } else {
        return var1 / var3;
    }
}
}
```

